

MCTTA: Multi-class Token Transformer Adapter for Weakly Supervised Semantic Segmentation

Dingjie PENG and Wataru KAMEYAMA

Abstract—Weakly Supervised Semantic Segmentation (WSSS) has emerged as a promising alternative to fully supervised methods, leveraging image-level labels instead of pixel-level annotations. In this paper, we systematically analyze the limitations of existing WSSS architectures, identifying their reliance on a single-structure design that lacks hierarchical feature extraction and multi-level feature fusion. In response to these shortcomings, we propose the Multi-class Token Transformer Adapter (MCTTA), which integrates graph-based multi-scale feature extraction as image prior and a hierarchical design into the MCTformer+ baseline. MCTTA introduces several key modules, including Spatial Prior Grapher (SPG), Class Token Projection (CTP), Cross Attention Block, and Split Weighted Softmax, along with a multi-scale feature fusion strategy inspired by Fully Supervised Semantic Segmentation (FSSS). These innovations significantly enhance Class Activation Map (CAM) generation and segmentation prediction, leading to substantial performance gains. Extensive experiments on PASCAL VOC 2012 and MS COCO 2014 datasets demonstrate the effectiveness of MCTTA with both direct and single-stage variants, achieving the state-of-the-art performance in single-stage WSSS. As well as, the multi-stage version of MCTTA outperforms the existing multi-stage methods. Furthermore, comprehensive ablation studies validate the effectiveness of each proposed module, showing consistent improvements across multiple configurations.

Index Terms—Adapter, class activation map, transformer, weakly supervised semantic segmentation.

I. INTRODUCTION

FULLY Supervised Semantic Segmentation (FSSS) aims to assign a semantic label to each pixel in an image, making it a fundamental task in computer vision with applications in autonomous driving [1], medical imaging [2], and scene understanding [3]. However, traditional FSSS models rely on large-scale datasets with manually annotated pixel-wise labels. While these models achieve high accuracy, the process of obtaining pixel-level annotations is labor-intensive and expensive. This challenge has motivated researchers to explore alternative approaches as Weakly Supervised Semantic Segmentation (WSSS) that reduce reliance on costly pixel-wise annotations. WSSS can utilize various types of supervision, including image-level labels [4], scribbles and bounding boxes [5], and points [6], significantly reducing annotation costs. Among these, image-level labels are the most accessible, as they only require indicating the presence of object categories in an image without specifying their exact locations. Despite its advantages, WSSS still faces several challenges. A key

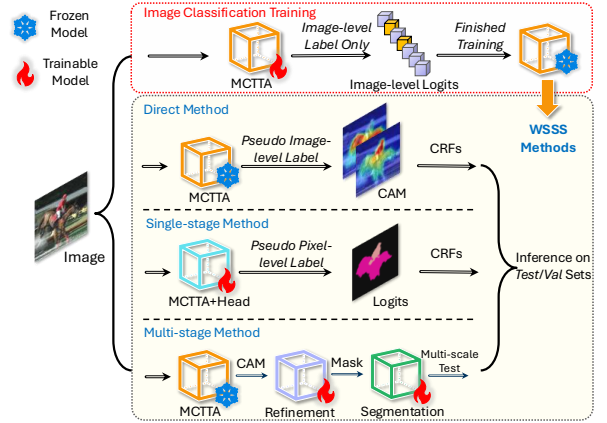


Fig. 1. Proposed MCTTA for weakly supervised semantic segmentation.

difficulty lies in generating high-quality pseudo labels from weak annotations, as methods relying on Class Activation Maps (CAMs) [7] often suffer from incomplete object localization and background noise. Additionally, existing WSSS frameworks, such as Convolutional Neural Network (CNN)-based models [8, 9], present challenges such as incomplete object localization and background contamination. CNNs tend to focus only on the most discriminative object parts due to the small effective receptive-field, leading to fragmented CAMs, while co-occurring contextual cues introduce noise, making it difficult to separate foreground from background. To address these issues, existing methods refine CAMs through region-growing techniques [10], pixel affinity propagation [11], and multi-stage learning [12] to expand object coverage. Additionally, contrastive learning and regularization losses help suppress noise and improve segmentation accuracy. These challenges necessitate the development of more effective strategies for refining CAMs under weak supervision.

Some proposals present other network architectures for vision tasks, such as Vision GNN (ViG) [13] and Vision Transformers (ViT) [14]. In ViG, an image is processed as a graph, where each node corresponds to a patch, and edges define the relationships between them. This structure allows ViG to capture irregular and long-range dependencies, making them particularly effective for tasks involving complex object shapes and scene layouts. On the other hand, ViT process images as a sequence of tokens, where an input image is divided into non-overlapping patches, that are flattened and embedded into feature vectors. The self-attention in ViT allows global feature interactions, making them highly effective in capturing long-range dependencies and improving contextual

Dingjie PENG is with Graduate School of Fundamental Science and Engineering, Waseda University, Tokyo, 169-8555 Japan (e-mail: kephier9013@asagi.waseda.jp).

Wataru KAMEYAMA is with Faculty of Science and Engineering, Waseda University, Tokyo, 169-8555 Japan (e-mail: wataru@waseda.jp).

reasoning. In WSSS, MCTformer [15] and MCTformer+ [16] induce a multi-class token manner inside the ViT, making the object localization maps more class-specific.

We argue that both ViG and ViT offer significant advantages over traditional CNNs by capturing global relationships and reducing locality bias. While ViG models structural dependencies through a graph-based framework, ViT leverages tokenized representations and self-attention for richer visual features. Building on these advancements, we introduce an adapter for MCTformer+ that fully integrates the strengths of ViG and ViT, named as Multi-class Token Transformer Adapter (MCTTA). Specifically, we propose Spatial Prior Grapher (SPG) to generate hierarchical features using a graph-based Grapher module. These multi-scale graph-like features are treated as hierarchical tokens, which, through Class Token Projection (CTP), refine multi-class representations and interact with backbone features via the Cross Attention (CAtnn) Block.

Notably, in MCTformer series [15, 16], we observe imbalance in attention scores caused by the smaller number of class tokens relative to patch tokens. To address this, we introduce split weighted softmax across all attention modules. Finally, following the standard practice in FSSS, we introduce hierarchical feature fusion to aggregate the final backbone layer output with multi-scale representations, to enhance classification prediction and CAM generation, ensuring a more comprehensive capture of global context and fine-grained spatial details.

For WSSS, our method leverages classification-pretrained MCTTA for flexible training and inference strategies, encompassing three primary approaches: the direct method, single-stage method, and multi-stage method, as depicted in Fig. 1.

In the direct method (MCTTA-D), the pretrained MCTTA generates CAMs as pseudo image-level labels, which are refined using Conditional Random Fields (CRFs) [17] for segmentation inference. This approach eliminates the need for additional segmentation network training, reducing computational cost.

The single-stage method (MCTTA-S) introduces a segmentation head trained alongside MCTTA. CAMs are refined into pseudo pixel-level labels, enabling end-to-end training of the segmentation network while maintaining efficiency and improving performance.

The multi-stage method (MCTTA-M) follows the conventional WSSS pipeline, where CAMs undergo further refinement through a dedicated refinement network. These refined masks serve as pseudo labels for training a segmentation network, achieving higher accuracy at the expense of increased training complexity.

By integrating these strategies, MCTTA provides a scalable and adaptable solution for WSSS, balancing computational efficiency and segmentation accuracy.

Our main contributions are as follows:

- We propose MCTTA, which integrates a graph-based architecture and hierarchical design into MCTformer+, providing a flexible and scalable solution for WSSS that balances efficiency and accuracy across different supervision levels.

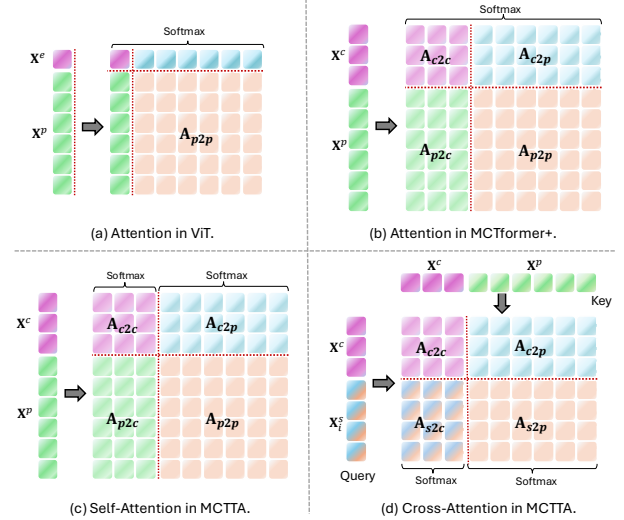


Fig. 2. Attention maps in (a): ViT [14], (b): MCTformer+ [16], and (c-d): MCTTA.

- We identify key limitations in existing WSSS architectures, and introduce carefully designed modules to address them. Through extensive ablation studies, we evaluate the contribution of each module, enabling MCTTA to achieve faster optimization and produce higher-quality CAMs.
- Our MCTTA-D performs on par with or exceeds existing single-stage methods. MCTTA-S achieves the best performance among all single-stage approaches, even surpassing some multi-stage methods. Meanwhile, MCTTA-M sets a new state-of-the-art in multi-stage WSSS, significantly enhancing segmentation performance.

This paper is organized as follows: Section II presents the related work pertinent to our study, including WSSS methods and vision adapters. Section III lays out the foundational preliminaries underlying our approach. Section IV provides a detailed description of the proposed MCTTA architecture and its method for WSSS, and Sections V and VI report the experimental results and the corresponding ablation studies, respectively. Finally, Section VII concludes the paper.

II. RELATED WORK

A. Weakly Supervised Semantic Segmentation

In WSSS, CAMs localize object regions by highlighting the most discriminative areas in an image based on a classification model's response. Many methods leverage CAMs with only image-level labels for training, typically categorized into multi-stage and single-stage methods.

Multi-stage Methods. Various techniques have been developed to generate more complete raw CAMs, focusing on cross-image semantics [18], ensemble method [19], erasing and searching strategies [20, 21], and optimization with additional losses [22, 23]. Some approaches also leverage feature correlation, including PPC [24], SIPE [25], and RS+EPM [26].

With the advent of Transformers, TSCAM [10] has introduced class-patch attention maps for improved localization, while the MCTformer series [15, 16] has enhanced feature

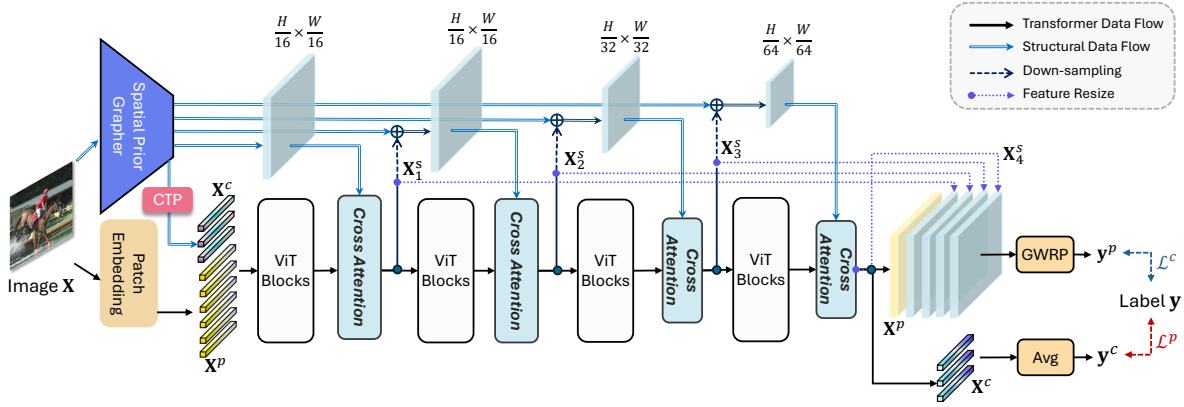


Fig. 3. The overall architecture of the proposed Multi-class Token Transformer Adapter (MCTTA).

discrimination through a class-specific token architecture. To refine CAMs into more accurate pseudo labels, PSA [11] and IRN [12] utilize pixel affinity and inter-pixel relations, while AMN [9], AdvCAM [27], and LPCAM [28] refine CAMs through threshold manipulation and spatial clustering. For segmentation, URN [34] reduces label noise by computing pixel-wise variance across prediction maps at different scales using cross-view consistency. ADELE [29] introduces adaptive annotation correction based on model outputs, while BECO [30] employs boundary-enhanced co-training to improve segmentation robustness.

Single-Stage Methods. Unlike multi-stage approaches, single-stage methods simplify the training process by learning an end-to-end semantic segmentation model supervised by image-level labels. Early works by Papandreou *et al.* [31] and Pinheiro *et al.* [32] have pioneered deep learning-based single-stage frameworks. Nikita and Stefan [33] have improved performance by enhancing local consistency and semantic fidelity, while Tang *et al.* [34] have introduced regularized losses to stabilize segmentation training. Zhang *et al.* [35] have proposed a dual-branch model to generate CAMs and refine confident yet compact regions. More recently, ToCo [36] has addressed the over-smoothing issue with a token contrast strategy, while DuPL [37] has introduced a dual-student framework to enhance progressive learning.

B. Adapters for Vision

In computer vision, various adaptation techniques have been developed to improve model transferability across tasks and domains. DAM [38] and SpotTune [39] focus on incremental learning and transfer learning, while domain adaptation methods like Residual-Adapter [40] and Vi-Adapter [41] dynamically adjust models for diverse visual domains. With the introduction of CLIP [42], numerous adapter-based approaches have emerged to transfer pretrained knowledge to zero-shot and few-shot tasks. Side-Adapter [43] formulates the task as a region recognition problem, making mask proposals CLIP-aware, while GBA [44] enhances the generalization and robustness of vision-language models.

For semantic segmentation, several adapter-based methods have been proposed. Click-Adapter offers a simple yet

effective interactive segmentation model without requiring pretraining. Domain-Rectifying Adapter [45] aligns diverse target domain styles with the source domain for improved segmentation. SegAdapter [46] leverages High-order Spatial Attention to integrate intermediate features from a pretrained backbone, enhancing segmentation performance. ViT-Adapter [47] incorporates image priors into the adapter, helping ViTs learn multi-domain representations from large-scale datasets.

III. PRELIMINARIES

A. MCTformer Series

As illustrated in Fig. 2(a), ViT [14] concatenates a single class token $\mathbf{X}^e$ with patch tokens $\mathbf{X}^p$ as input for self-attention, resulting in a class-agnostic attention map that lacks specificity in object localization.

To overcome this limitation, MCTformer [15] introduces a multiple-class token scheme within ViT, enabling the model to capture class-specific attention for more precise object localization. MCTformer processes an image $\mathbf{X} \in \mathbb{R}^{3 \times H \times W}$ by dividing it into N non-overlapping patches, which are then flattened to form patch tokens $\mathbf{X}^p \in \mathbb{R}^{N \times D}$, where H and W denote the height and width of the image, N represents the number of patches, and D denotes the embedding dimension. A single class token $\mathbf{X}^e \in \mathbb{R}^{1 \times D}$ is expanded into a multi-class scheme by duplicating it N_c times, forming multi-class tokens $\mathbf{X}^c \in \mathbb{R}^{N_c \times D}$, where N_c is the number of categories. As shown in Fig. 2(b), within the ViT blocks, $\mathbf{X}^c$ is concatenated with $\mathbf{X}^p$ and processed through multiple transformer layers. Global Average Pooling (GAP) is then applied to the final feature maps to generate classification predictions $\mathbf{y}^p \in \mathbb{R}^{N_c}$, while the multi-class tokens $\mathbf{X}^c$ produce an additional class label prediction $\mathbf{y}^c \in \mathbb{R}^{N_c}$, as shown in Fig. 3.

To further enhance CAM quality, MCTformer+ [16] replaces GAP with Global Weighted Ranking Pooling (GWRP) [48] and introduces Contrastive Class Tokens (CCT) loss, improving class-token learning for more discriminative representations.

B. Vision GNN

Vision GNN (ViG) [13] models images as graph structures to effectively capture diverse and intricate object shapes

through the Grapher module, which employs graph convolution to aggregate and update graph information. Similar to MCTformer, the input to the Grapher consists of patch tokens $\mathbf{X}^p = [\mathbf{x}_1, \dots, \mathbf{x}_N]$ with a dimension of D . These tokens are treated as an unordered set of nodes. For each node $\mathbf{x}_i$, the Grapher identifies its k -nearest neighbors using a max relation and establishes directed edges e_{ji} from $\mathbf{x}_j$ to $\mathbf{x}_i$ for all $\mathbf{x}_j \in \mathcal{N}(\mathbf{x}_i)$. Denoting the graph construction process as $G(\mathbf{X})$, the Grapher operates as:

$$G(\mathbf{x}_i) = ([\mathbf{x}_i, \max(\mathbf{x}_j - \mathbf{x}_i | j \in \mathcal{N}(\mathbf{x}_i))] \mathbf{W}_a) \mathbf{W}_u, \quad (1)$$

where $\mathbf{W}_{a,u} \in \mathbb{R}^{D \times D}$ are the weights for aggregation and updating, respectively.

IV. PROPOSED METHODS

A. Overview

In this section, we delve into the details of the proposed MCTTA architecture and our WSSS methods. We first begin with the limitation of the existing WSSS architectures in Section IV-B, then we introduce the architecture of MCTTA in Section IV-C in detail, including the spatial prior grapher, class token projection, cross attention, and split weighted softmax. The WSSS methods are discussed in Section IV-D, including the multi-stage, single-stage and direct method.

B. Limitations in WSSS Architectures

Although Transformer-based approaches like MCTformer+ [16] and ToCo [36] generate high-quality CAMs, we identify several limitations in their architectures for WSSS:

- **Lack of hierarchical feature extraction.** Models in WSSS, including MCTformer+, often use an isotropic architecture that processes features at a single level. However, dense prediction tasks such as segmentation benefit from multi-level hierarchical features to capture fine details.
- **Single-structure limitations.** Existing methods rely solely on either Transformer or CNN architectures, limiting their ability to model both long-range dependencies and fine-grained boundary information.
- **Mismatch between FSSS and WSSS feature utilization.** FSSS relies on hierarchical features from shallow to deep layers, whereas WSSS typically uses only the final feature map. While sufficient for classification, this is suboptimal for generating high-quality CAMs.

To address these challenges, we propose an improved architecture that enhances feature extraction and segmentation quality by integrating hierarchical representations and leveraging the strengths of both ViG and ViT. Inspired by FSSS design, we adopt a simple fusion strategy to fully utilize hierarchical features along with the final backbone layer output to bridge the gap between FSSS and WSSS.

C. MCTTA Architecture

As illustrated in Fig. 3, MCTTA is built upon the ViT. The model takes an image $\mathbf{X} \in \mathbb{R}^{3 \times H \times W}$ as input, and utilizes the corresponding class label $\mathbf{y} \in \mathbb{R}^{N_c}$ as the objective.

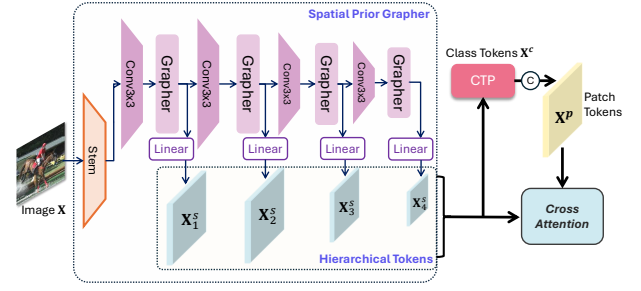


Fig. 4. The architecture of the proposed Spatial Prior Grapher (SPG).

Through the proposed Spatial Prior Grapher (SPG), Class Token Projection (CTP), and Cross Attention (CAtn) Block, it produces two predictions: $\mathbf{y}^p \in \mathbb{R}^{N_c}$ and $\mathbf{y}^c \in \mathbb{R}^{N_c}$. We divide the MCTTA data flow into four steps:

- 1) **Patch/Hierarchical Embedding:** the image $\mathbf{X}$ is split into N non-overlapped patches, forming as patch tokens $\mathbf{X}^p$. Besides, in adapter branch, $\mathbf{X}$ is also proceeded with the SPG to generate a set of hierarchical tokens as $\{\mathbf{X}_i^s\}$, where i denotes the i -th stage in adapter.
- 2) **Multi-Class Token Generation:** the proposed CTP is applied to produce the multi-class token $\mathbf{X}^c$ based on the combination of hierarchical tokens $\{\mathbf{X}_i^s\}_{i=1}^S$ and the single class tokens $\mathbf{X}^e$ from the DeiT [49].
- 3) **Patch/Hierarchical Process:** patch tokens $\mathbf{X}^p$ and multi-class token $\mathbf{X}^c$ are concatenated together to go through L ViT blocks. Meanwhile, hierarchical tokens $\mathbf{X}^s$ are interacted with the $\mathbf{X}^p$ and $\mathbf{X}^c$ via the CAtn Block.
- 4) **Logits Prediction:** the final patch tokens $\mathbf{X}^p$ and hierarchical tokens $\{\mathbf{X}_i^s\}$ are concatenated and aggregated by GWRP to produce class logits $\mathbf{y}^p$, while the final multi-class token $\mathbf{X}^c$ is averaged along the embedded dimension for logits prediction $\mathbf{y}^c$.

1) *Spatial Prior Grapher:* We propose a novel Spatial Prior Grapher (SPG) for hierarchical feature extraction. As depicted in Fig. 4, at each hierarchical level i , SPG first applies simple convolution stem (denoted as Stem) to down-sample $\mathbf{X}$ with a stride of 4. Then a sequential of 3×3 convolution (denoted as Conv $_{3 \times 3}$) and Grapher module (denoted as Grapher) [13] is employed to produce a spatial feature $\bar{\mathbf{X}}_i^s$. By denoting D_i as the channel dimension for each level i , and $H_i \times W_i$ the size of i -th hierarchical tokens, we adopt a linear transformation to project $\bar{\mathbf{X}}_i^s \in \mathbb{R}^{D_i \times H_i \times W_i}$ to $\mathbf{X}_i^s \in \mathbb{R}^{D \times H_i \times W_i}$ to align the feature dimension in ViT:

$$\begin{aligned} \bar{\mathbf{X}}_0^s &= \text{Stem}(\mathbf{X}), \\ \bar{\mathbf{X}}_{i+1}^s &= \text{Grapher}(\text{Conv}_{3 \times 3}(\bar{\mathbf{X}}_i^s)). \\ \mathbf{X}_{i+1}^s &= \text{Linear}(\bar{\mathbf{X}}_{i+1}^s). \end{aligned} \quad (2)$$

By adjusting the stride and padding of the convolution, the size of the downsampled hierarchical features can be effectively controlled. The effects of the stride and the Grapher module are discussed in Section VI-B and VI-C, respectively.

2) *Class Token Projection:* Since CAM generation heavily relies on class-discriminative semantics, achieving rapid converge in multi-class token prediction is crucial for effective classification training [16]. Under the experimental conditions

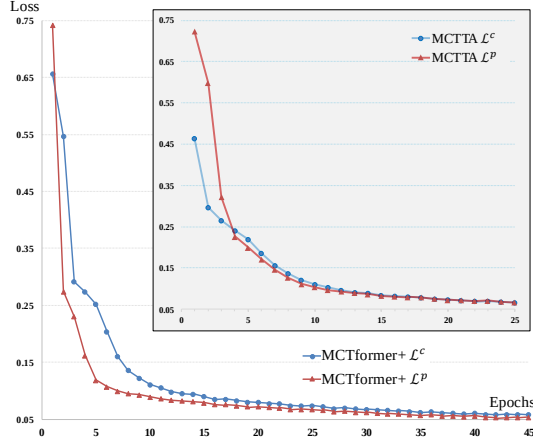


Fig. 5. The losses of the MCTformer+ and the proposed MCTTA for the class token prediction loss $\mathcal{L}^c$ and (fused) patch prediction loss $\mathcal{L}^p$.

provided in Section V-B, with the same trading off parameter $\lambda = 1$, Fig. 5 shows that the multi-class token prediction loss $\mathcal{L}^c$ for MCTformer+ is significantly higher than the patch token prediction loss $\mathcal{L}^p$ during the initial epochs on VOC train set [50], which indicate a slower learning speed for class-discriminative information in $\mathbf{X}^c$. We attribute this to the initialization of the multi-class token $\mathbf{X}^c \in \mathbb{R}^{N_c \times D}$ through simple duplication of the single class token $\mathbf{X}^e \in \mathbb{R}^{1 \times D}$ from DeiT [49], which is based on class-agnostic aggregation. Consequently, $\mathbf{X}^c$ can only learn class-specific semantics indirectly through self-attention.

To address this limitation, we propose the Class Token Projection (CTP) method to enhance the training of $\mathbf{X}^c$. As shown in Fig. 6, the hierarchical tokens $\{\mathbf{X}_i^s\}$ form a set of multi-scale graph-like features, capturing semantic information from shallow to deep layers. Each multi-level feature is processed by GWRP to generate a class token $\mathbf{X}^{e'} \in \mathbb{R}^{1 \times D}$. These tokens are concatenated along the embedding dimension (denoted as $\text{Cat}(\cdot)$) and passed through a linear transformation with weights $\mathbf{W}_c \in \mathbb{R}^{S \times N_c}$, resulting in an aggregated token $\mathbf{X}^{e'} \in \mathbb{R}^{N_c \times C}$. Finally, by adding the duplication of $\mathbf{X}^e$ to $\mathbf{X}^{e'}$, we obtain the multi-class token $\mathbf{X}^c$ for MCTTA. This process can be expressed as follows:

$$\begin{aligned} \mathbf{X}^{e'} &= \text{Cat}(\text{GWRP}(\{\mathbf{X}_i^s\}))\mathbf{W}_c, \\ \mathbf{X}^c &= \text{Dup}(\mathbf{X}^e) + \mathbf{X}^{e'}, \end{aligned} \quad (3)$$

where $\text{Dup}(\cdot)$ extends the shape of $\mathbf{X}^e$ from $1 \times D$ to $N_c \times D$. As shown in Fig. 5, similar to MCTformer+, the trade-off weight between $\mathcal{L}^c$ and $\mathcal{L}^p$ is set to $\lambda = 1$ for a fair comparison. However, in our experiments, we use $\lambda = 3$ as the default setting.

From Fig. 5, the proposed CTP significantly reduces the initial prediction loss $\mathcal{L}^c$ in MCTTA, enabling faster convergence compared to MCTformer+. Specifically, on the VOC dataset, MCTTA converges within 25 epochs, whereas MCTformer+ requires 45 epochs. The impact of CTP is further analyzed in Section VI-E.

3) *Cross Attention Block*: We propose the Cross Attention (CAtn) Block to facilitate interaction among multi-class to-

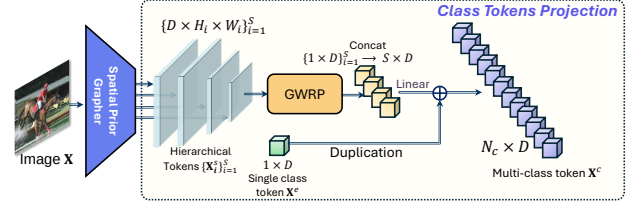


Fig. 6. Illustration of the proposed Class Token Projection.

kens $\mathbf{X}^c$, patch tokens $\mathbf{X}^p$, and hierarchical tokens $\{\mathbf{X}_i^s\}$. Specially, the L MCTformer blocks are evenly divided into $S = 4$ parts for interaction with hierarchical tokens. In essence, the final multi-class tokens $\mathbf{X}^c$ and patch tokens $\mathbf{X}^p$ of each part are concatenated and processed with Layer Normalization (LN), then projected as the key $\mathbf{K}$ and value $\mathbf{V}$. Simultaneously, the hierarchical token $\mathbf{X}_i^s$ is concatenated with $\mathbf{X}^c$, and then projected as the query $\mathbf{Q}$ similarly. Since $\{\mathbf{X}_i^s\}$ is a 2D grid form, which differs from that of $\mathbf{X}^p$, a sequence 1D form, we reshape $\{\mathbf{X}_i^s\}$ into a sequence of tokens with dimensions $N_i \times D$, where $N_i = H_i \times W_i$. The cross-attention process in each hierarchical level i is defined as follows:

$$\begin{aligned} \mathbf{Q} &= (\text{LN}(\text{Cat}(\mathbf{X}_i^s, \mathbf{X}^c)))\mathbf{W}_q, \\ \mathbf{K} &= (\text{LN}(\text{Cat}(\mathbf{X}^p, \mathbf{X}^c)))\mathbf{W}_k, \\ \mathbf{V} &= (\text{LN}(\text{Cat}(\mathbf{X}^p, \mathbf{X}^c)))\mathbf{W}_v, \\ \mathbf{Z} &= \mathbf{A}\mathbf{V} = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{D}}\right)\mathbf{V}, \end{aligned} \quad (4)$$

where $\mathbf{W}_{q,k,v} \in \mathbb{R}^{D \times D}$ are the linear transformation weights, $\text{Cat}(\cdot)$ denotes token concatenation, $\text{LN}(\cdot)$ applies layer normalization, and $\mathbf{Z} \in \mathbb{R}^{(N_c + N_i) \times D}$ represents the output of CAtn.

In experiments, we employ a multi-head scheme for the cross attention and include a feed-forward network (FFN) as a subsequent step. The CAtn Block can be expressed as:

$$\begin{aligned} \mathbf{Z}' &= \mathbf{X} + \text{CAtn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}), \\ \mathbf{Z} &= \mathbf{Z}' + \text{FFN}(\text{LN}(\mathbf{Z}')). \end{aligned} \quad (5)$$

After processing through the CAtn Block, the output $\mathbf{Z}$ is split into a new multi-class tokens $\mathbf{Z}[:, N_c]$ and a new hierarchical token as $\mathbf{Z}[N_c :]$. Here, $[:, N_c]$ and $[N_c :]$ denote the slicing function that select the first N_c and left behind elements in $\mathbf{Z}$, respectively.

Then, $\mathbf{Z}[N_c :]$ is downsampled to match the size of the next level token $\mathbf{X}_{i+1}^s$ and added for updating:

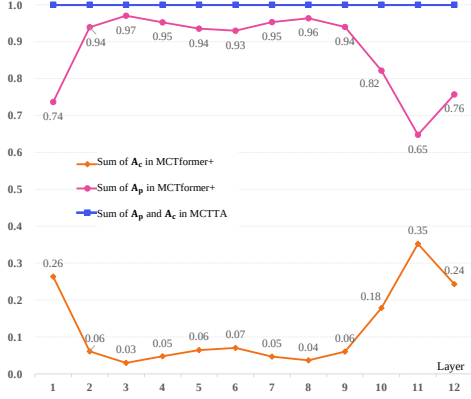
$$\mathbf{X}_{i+1}^s \leftarrow \text{DS}(\mathbf{Z}[N_c :]) + \mathbf{X}_{i+1}^s, \quad (6)$$

where $\text{DS}(\cdot)$ denotes the downsampling operation.

Additionally, the multi-class tokens are updated using a zero-initialized weight $\mathbf{w}_z \in \mathbb{R}^D$ as:

$$\mathbf{X}^c \leftarrow \mathbf{w}_z \odot \mathbf{Z}[:, N_c] + \mathbf{X}^c. \quad (7)$$

4) *Split Weighted Softmax*: Vanilla self-attention applies softmax to normalize attention scores, so that the sum of all columns equals 1. However, in MCTformer+, self-attention is computed as $\mathbf{A} = \text{Softmax}(\mathbf{Q}\mathbf{K}^\top/\sqrt{D})$, where the multi-class token $\mathbf{X}^c$ and patch tokens $\mathbf{X}^p$ are concatenated to project query, key and value. Since the number of classes

Fig. 7. Sum of the attention scores for A_c and A_p per layer.

N_c is typically much smaller than the number of patches N (e.g., $N_c = 20, N = 784$ in VOC [50]), vanilla softmax causes the probability densities to be heavily concentrated on the patch tokens.

As shown in Fig. 7, following the experimental condition described in Section V-B, we plot the sum of attention scores for the multi-class tokens $A_c = A[:, : N_c]$ and patch tokens $A_p = A[:, N_c :]$ across all 12 layers on the VOC train set, where $[:, : N_c]$ denotes the slicing function that select the first N_c columns in A_c . Notably, the weights in A_c tend to become larger in the last three layers, probably because the loss $\mathcal{L}^c$ is supervised on the final layer class token. However, except for the first and last few layers, the sum of A_p exceeds 0.9, while the highest sum of A_c is only 0.35. The heat map shown in Fig. 8(a) also reveals that all columns in the MCTformer+ attention map A get relatively small activations, including the part of A_c .

To address this, we propose split weighted softmax, which applies softmax separately with weighting factors α_1, α_2 to A_c and A_p :

$$\begin{aligned} A'_c &= \alpha_1 \cdot \text{Softmax}(A[:, : N_c]), \\ A'_p &= \alpha_2 \cdot \text{Softmax}(A[:, N_c :]), \\ A &\leftarrow \text{Cat}(A'_c, A'_p). \end{aligned} \quad (8)$$

By adjusting the weights $\alpha = (\alpha_1, \alpha_2)$, we can balance the attention distribution between the multi-class tokens and patch tokens. In default, we set $\alpha_1 = 1$ and $\alpha_2 = 1$ since we regard the multi-class tokens and patch tokens contribute equally to the final prediction. As shown in Fig. 7 and Fig. 8(b), the attention module can obtain more balanced information from the class-token and patch tokens.

Additionally, as depicted in Fig. 2, in the proposed CAttn Block, the query (key) is also derived from the concatenation of X^c and X^p (X_i^s), which forms four attention parts as class-to-class A_{c2c} , class-to-patch A_{c2p} , spatial-to-class A_{s2c} and spatial-to-patch A_{s2p} . Therefore, split weighted softmax is also applied in CAttn. The effectiveness of this approach is further analyzed in Section VI-F.

5) *Hierarchical Features Fusion*: In FSSS, features from all hierarchical levels are fused to produce the final dense prediction, as seen in models like UNet [2], UpperNet [51],

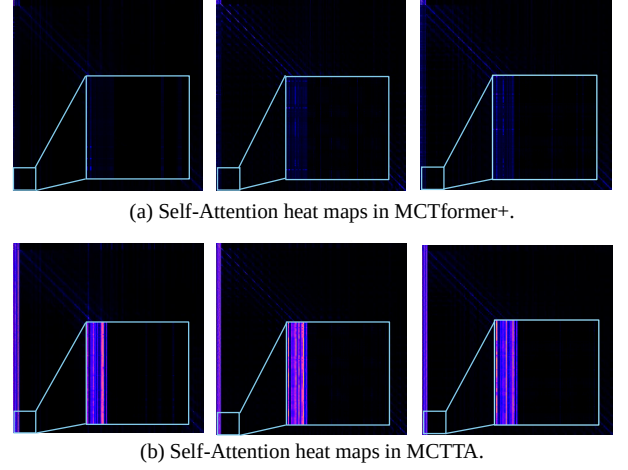


Fig. 8. Self-Attention heat maps in MCTformer+ [16] and MCTTA.

and FCN [52], etc. Inspired by this, we fuse all hierarchical levels $\{X_i^s\}$ from the adapter along with the patch tokens X^p from the backbone, resulting in a unified feature representation for both image-level label prediction and CAM generation. Specifically, all $\{X_i^s\}$ are resized to match the size of X^p , then concatenated. This fused representation is processed through a simple 1×1 convolution block (denoted as $\text{Conv}_{1 \times 1}$), reducing the channel dimensions from $5D$ to D :

$$\begin{aligned} X^f &= \text{Cat}(\text{Resize}(\{X_i^s\}_{i=1}^S, X^p)) \\ X^{p'} &= \text{Conv}_{1 \times 1}(X^f). \end{aligned} \quad (9)$$

The effect of the hierarchical levels is discussed further in Section VI-D.

6) *Classification Training Objectives*: In classification, MCTTA generates two logits for image-level label prediction: y^c from the multi-class token X^c , and y^p from the fused features $X^{p'}$. The training objective is defined as:

$$\mathcal{L} = \lambda \cdot \mathcal{L}^c + \mathcal{L}^p = \lambda \cdot \mathcal{L}(y, y^c) + \mathcal{L}(y, y^p), \quad (10)$$

where $y^c = \text{GWRP}(X^c)$ and $y^p = \text{GWRP}(X^{p'})$. Here, $\mathcal{L}(y, y^c)$ and $\mathcal{L}(y, y^p)$ represent the multi-label soft margin loss for class token and fused patch token predictions, respectively. Empirically, increasing λ accelerates the convergence of y^c , improving overall performance. We set $\lambda = 3.0$ as the default trade-off weight, with further analysis provided in Section VI-G.

7) *Class Activation Maps Generation*: We follow the standard settings as in [15, 16] to obtain the non-normalized CAM (denoted as CAM^p), based on the feature representation $X^{p'}$, which is computed as:

$$\text{CAM}^p = \text{ReLU}(\mathbf{W}_h^T X^{p'}) \in \mathbb{R}^{N_c \times \frac{H}{16} \times \frac{W}{16}}, \quad (11)$$

where $\mathbf{W}_h$ represents the weights of the prediction head, and $\text{ReLU}(x) = \max(0, x)$ is a non-linear activation function that filters out negative scores in CAM^p .

For class-specific CAM refinement, we adopt the same process as in MCTformer+: attention maps from the last L_c transformer layers are aggregated and extracted as class-to-

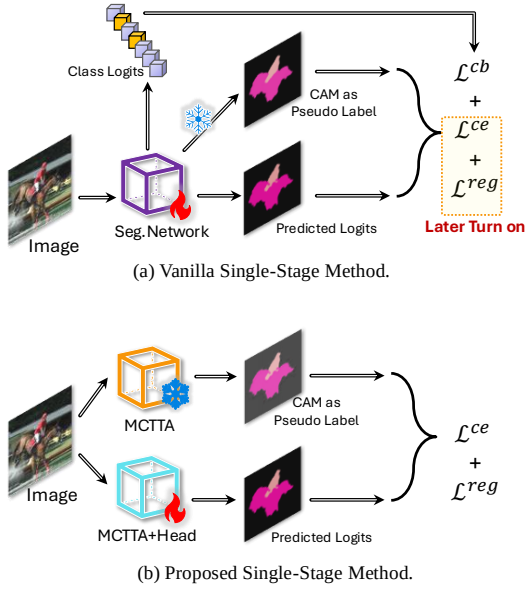


Fig. 9. Illustrations of the vanilla and proposed single-stage method.

patch attention $\mathbf{A}_{c2p} \in \mathbb{R}^{N_c \times N}$ to refine $\mathbf{CAM}^p$. Specifically, the refinement is formulated as:

$$\mathbf{A}_{c2p} = \prod_{l=L-L_c+1}^L \mathbf{A}^l[:, N_c, N_c :], \quad (12)$$

$$\mathbf{CAM}^u = \text{Flatten}(\mathbf{CAM}^p) \otimes \mathbf{A}_{c2p} \in \mathbb{R}^{N_c \times N},$$

where $\text{Flatten}(\cdot)$ reshapes $\mathbf{CAM}^p$ into dimensions $N_c \times N$, and $\otimes$ denotes element-wise multiplication for feature selection.

Next, the patch-to-patch attention $\mathbf{A}_{p2p} \in \mathbb{R}^{N \times N}$, derived from the last L_p layers, is applied to model patch-to-patch affinities and obtain the final CAM:

$$\mathbf{A}_{p2p} = \sum_{l=L-L_p+1}^L \mathbf{A}^l[N_c :, N_c :] \quad (13)$$

$$\mathbf{CAM}_{c(j)} = \sum_k \mathbf{A}_{p2p}(j, k) \cdot \mathbf{CAM}_{c(k)}^u,$$

where $\mathbf{CAM}_{c(j)}$ and $\mathbf{CAM}_{c(k)}^u$ denote the j -th and k -th patches of class c in $\mathbf{CAM}$ and $\mathbf{CAM}^u$, respectively. Following MCTformer+, we set $L_c = 3$ and $L_p = 12$ by default.

D. Methods for WSSS

1) *Multi-Stage Method*: We incorporate the proposed MCTTA into a multi-stage framework. As illustrated in Fig. 1, MCTTA is first trained for image classification using only image-level labels. Subsequently, the model generates CAMs, referred to as seeds, as detailed in Section IV-C7. These CAMs are then refined using by a refinement network, to create masks that act as pseudo pixel-level labels for segmentation training.

2) *Single-Stage Method*: To further validate the effectiveness of MCTTA, we integrate it into our single-stage method. As illustrated in Fig. 9(a), traditional single-stage methods begin with classification training using the loss function $\mathcal{L}^{cb}$ during the initial warm-up phase. This is followed by end-to-end segmentation training, where pseudo pixel-level labels are generated from a frozen network and optimized with cross-entropy loss $\mathcal{L}^{ce}$. Additionally, a regularization loss $\mathcal{L}^{reg}$ is applied to improve training stability.

TABLE I
COMPARISON OF THE mIoU (%) FOR THE RAW CAMS (SEED) AND THE PSEUDO LABELS (MASK) ON VOC AND COCO FOR MULTI-STAGE METHODS.

Method	VOC Train Seed	VOC Train Mask	COCO Train Seed	COCO Train Mask
PSA (CVPR18) [11]	48.0	61.0	-	-
IRN (CVPR19) [12]	48.8	66.5	33.1	42.5
SC-CAM (CVPR20) [53]	50.9	63.4	-	-
SEAM (CVPR20) [54]	55.4	63.6	25.1	31.5
CONTA (NeurIPS20) [55]	48.8	67.9	28.7	35.2
CDA (ICCV21) [56]	58.4	66.4	-	-
RIB (NeurIPS21) [57]	56.5	70.6	36.5	45.6
CLIMS (CVPR22) [58]	56.6	70.5	-	-
SIPE (CVPR22) [25]	58.6	64.7	-	-
AMN (CVPR22) [9]	62.1	72.2	40.3	46.7
W-OoD (CVPR22) [59]	59.1	72.1	-	-
MCTformer (CVPR22) [15]	61.7	69.1	36.6	41.6
ReCAM (CVPR22) [22]	54.8	70.9	37.8	46.3
LPCAM (CVPR23) [28]	65.3	72.7	42.5	47.7
ACR (ICCV23) [60]	60.3	72.3	-	-
SFC (AAAI24) [61]	64.7	73.7	-	-
CTI (CVPR24) [62]	69.5	74.7	-	-
MCTformer+ (TPAMI24) [16]	68.8	76.2	42.8	48.1
MCTTA-Cls (Ours)	73.9	82.3	46.3	52.2

However, this multi-task learning framework is challenging to optimize due to the low quality of pseudo labels, which can negatively impact segmentation performance.

In contrast, our approach, illustrated in Fig. 9(b), leverages frozen pretrained weights from classification to generate pseudo labels. The segmentation network, built on MCTTA, is then trained solely to produce dense prediction logits. Despite this separation, our method still qualifies as a single-stage approach, as utilizing pretrained weights is a common practice in [33, 36, 63]. We lists the merits of the proposed approach as follows:

- 1) Higher-quality pseudo masks. The pretrained weights benefit from a larger batch size during classification, enabling faster convergence and generating more reliable pseudo masks.
- 2) Simplified optimization. Traditional methods require joint optimization of classification and segmentation losses, whereas our approach focuses solely on dense prediction logits, making training more stable and efficient.

A detailed comparison between our proposed single-stage method and vanilla approach is provided in Section VI-A.

3) *Direct Method*: We propose a direct method that relies solely on classification training weights to generate masks for the val and test sets, eliminating the need for an additional segmentation network. This approach significantly reduces computational cost while maintaining high-quality mask generation. The CAM generation follows the same settings as described in Section IV-C7.

Since ground truth image labels $\mathbf{y}$ are unavailable for filtering out non-target categories, we use the class token prediction $\mathbf{y}^c$ as the pseudo label. To enhance robustness, we incorporate multi-scale flip testing, a common practice for CAM generation. Specifically, we generate an additional pseudo label $\bar{\mathbf{y}}^c$ from a horizontally flipped image. We then

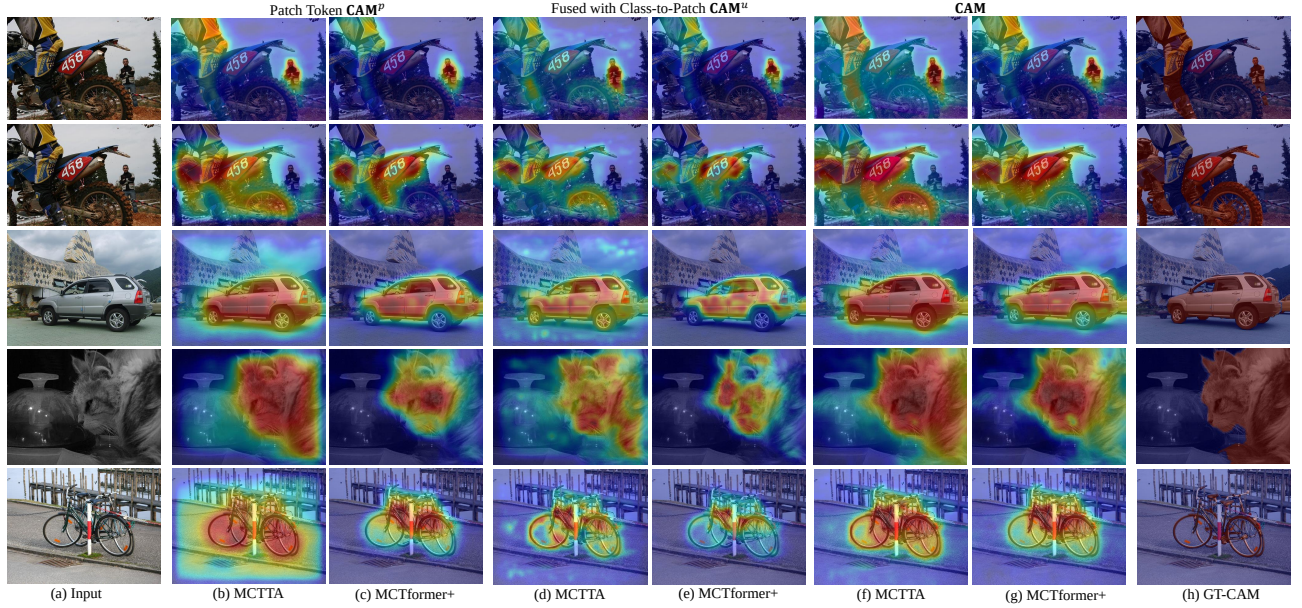


Fig. 10. Qualitative comparison of CAM results across different methods on VOC train set [50]. (a): input image. (b), (d), (f): patch token CAM^P , class-to-patch fused CAM^u , and final refined CAM from the proposed MCTTA. (c), (e), (g): corresponding results from MCTformer+ [16]. (h): ground truth CAM.

apply a logical AND operation (denoted as $\wedge$) to obtain a one-hot filter $\hat{\mathbf{y}}^c \in \mathbb{R}^{N_c}$ for non-target class suppression, producing the final prediction:

$$\begin{aligned} \hat{\mathbf{y}}^c &= \text{OneHot}(\mathbf{y}^c) \wedge \text{OneHot}(\bar{\mathbf{y}}^c) \\ \mathbf{M} &= \hat{\mathbf{y}}^c \odot \text{CAM}, \end{aligned} \quad (14)$$

where $\text{OneHot}(\cdot)$ converts logits into one-hot image-level labels, assigning 0 to non-positive values and 1 to positive values, and $\odot$ denotes element-wise multiplication along the channel dimension.

V. EXPERIMENTS

A. Datasets and Metrics

Our experiments are conducted on two widely used public datasets: PASCAL VOC 2012 (VOC) [50] and MS COCO 2014 (COCO) [64], both of which provide pixel-level annotations for segmentation. VOC includes 21 object categories (including background), while COCO contains 81. The VOC dataset consists of 1,464 training (train), 1,449 validation (val), and 1,456 test (test) images. Following the standard practice, we augment the train set using additional data as in [9, 27, 28] to obtain a total of 10,582 training images. The COCO dataset comprises 80k and 40k images for training and validation, respectively. In all experiments, we report the mean Intersection over Union (mIoU) for CAMs, pseudo labels and segmentation results.

B. Implementation Details

1) *Image Classification*: The proposed MCTTA is initialized with DeiT-S [49] pretrained on ImageNet-1k [65]. We apply standard data augmentation techniques, including random horizontal flipping, color jittering, scaling, and cropping to a fixed image size of 448×448 .

The hierarchical embedding dimensions are set to $D_i = \{192, 384, 384, 768\}$, with an output stride configuration of $[16, 16, 32, 64]$ across four hierarchical levels. The model is trained using the AdamW optimizer [66] with a learning rate of 5×10^{-4} and a batch size of 32. We train for 25 epochs on VOC and 40 epochs on COCO.

2) *Multi-Stage Method Settings*: To ensure a fair comparison, we follow the CAM refinement strategies from previous works [9, 12, 28, 30]. Raw CAMs (seeds) are generated using a multi-scale setting of (0.75, 1.0, 1.25), and IRN [12] is used for CAM refinement to produce high-quality pseudo masks.

For segmentation, we use the MMSegmentation framework [67] with a consistent training and testing pipeline at an input size of 512×512 . We adopt VOC and COCO pretrained ResNet-101 [8] as the backbone, integrated with DeepLab v3+ [68]. The model is trained with the AdamW optimizer, using a batch size of 16 and a learning rate of 6×10^{-5} . DeepLab v3+ is trained for 40k iterations on VOC and 160k on COCO. Given the high quality of the generated pseudo masks, we apply multi-scale testing with scales (0.5, 0.75, 1.0, 1.25, 1.5) instead of using CRFs [17].

3) *Single-Stage Method Settings*: We construct the segmentation network using MCTTA as the backbone and the same segmentation head as in [36, 63]. The pretrained classification weights on VOC and COCO are used to generate pseudo labels and train the single-stage MCTTA with an input size of 448×448 .

To improve low-level semantic boundary alignment, pseudo labels are refined using Pixel Adaptive Refinement (PAR) [63]. The segmentation network is trained with a combination of cross-entropy loss $\mathcal{L}^{ce}$ and regularization loss $\mathcal{L}^{reg}$ [84]:

$$\mathcal{L}^s = \lambda_1 \cdot \mathcal{L}^{ce} + \lambda_2 \cdot \mathcal{L}^{reg}, \quad (15)$$

where λ_1 and λ_2 control the trade-off. Following ToCo [36], we set $\lambda_1 = 4$ and $\lambda_2 = 1$. The network is trained using the

TABLE II
COMPARISON OF THE mIoU (%) FOR THE SEGMENTATION RESULTS ON VOC VAL AND TEST FOR MULTI-STAGE METHODS.

Multi-stage Methods	Seg.Network	Sup.	Val	Test
EDAM (CVPR21) [69]	R101-V2	$\mathcal{I} + \mathcal{S}$	70.9	70.6
EPS (CVPR21) [70]	R101-V2	$\mathcal{I} + \mathcal{S}$	71.0	71.8
Yao <i>et al.</i> (CVPR21) [71]	R101-V2	$\mathcal{I} + \mathcal{S}$	68.3	68.5
AuxSegNet (ICCV21) [72]	WR38-V1	$\mathcal{I} + \mathcal{S}$	69.0	68.6
Du <i>et al.</i> (CVPR22) [73]	R101-V2	$\mathcal{I} + \mathcal{S}$	72.6	73.6
L2G (CVPR22) [74]	WR38-V1	$\mathcal{I} + \mathcal{S}$	72.0	73.0
RCA (CVPR22) [75]	WR38-V2	$\mathcal{I} + \mathcal{S}$	72.2	72.8
CLIMS (CVPR22) [58]	R101-V2	$\mathcal{I} + \mathcal{L}$	70.4	70.0
CLIP-ES (CVPR23) [76]	R101-V2	$\mathcal{I} + \mathcal{L}$	72.2	72.8
CLIP-ES+CPAL (CVPR24) [77]	R101-V2	$\mathcal{I} + \mathcal{L}$	74.5	74.7
SC-CAM (CVPR2020) [53]	R101-V2	$\mathcal{I}$	66.1	65.9
RIB (NeurIPS21) [57]	R101-V2	$\mathcal{I}$	70.2	70.0
CDA (ICCV21) [56]	WR38-V1	$\mathcal{I}$	66.2	66.8
ESOL (NeurIPS22) [78]	R101-V2	$\mathcal{I}$	69.9	69.3
BES (ECCV20) [79]	R101-V2	$\mathcal{I}$	65.7	66.6
AdvCAM (CVPR21) [27]	R101-V2	$\mathcal{I}$	68.1	68.0
ECS-Net (ICCV21) [80]	WR38-V1	$\mathcal{I}$	66.6	67.6
AMN (CVPR22) [9]	R101-V2	$\mathcal{I}$	70.7	70.6
W-OoD (CVPR22) [59]	WR38-V1	$\mathcal{I}$	70.7	70.1
SIPE (CVPR22) [25]	WR38-V1	$\mathcal{I}$	68.2	69.7
ReCAM (CVPR22) [22]	R101-V2	$\mathcal{I}$	68.4	68.2
MCTformer (CVPR22) [15]	WR38-V1	$\mathcal{I}$	71.9	71.6
OCR (CVPR23) [81]	WR38-V1	$\mathcal{I}$	72.7	72.0
LPCAM (CVPR23) [28]	WR38-V1	$\mathcal{I}$	72.6	72.4
BECO (CVPR23) [30]	R101-V3+	$\mathcal{I}$	72.1	71.8
ACR (ICCV23) [60]	WR38-V1	$\mathcal{I}$	71.9	71.9
USAGE (ICCV23) [82]	WR38-V1	$\mathcal{I}$	71.9	72.8
CTI (CVPR24) [62]	WR38-V1	$\mathcal{I}$	74.1	73.2
AMN+CPAL (CVPR24) [77]	R101-V2	$\mathcal{I}$	72.5	72.9
MCTformer+ (TPAMI24) [16]	WR38-V1	$\mathcal{I}$	74.0	73.6
MCTTA-M (Ours)	R101-V3+	$\mathcal{I}$	77.8	77.5

$\mathcal{I}$: Image level label, $\mathcal{L}$: language label, $\mathcal{S}$: Saliency label.

TABLE III
COMPARISON OF THE mIoU (%) FOR THE SEGMENTATION RESULTS ON COCO VAL FOR MULTI-STAGE METHODS.

Multi-stage Methods	Seg. Network	Sup.	Val
EPS (CVPR21) [70]	VGG16-V2	$\mathcal{I} + \mathcal{S}$	35.7
RCA (CVPR22) [75]	VGG16-V2	$\mathcal{I} + \mathcal{S}$	36.8
AuxSegNet (ICCV21) [72]	WR38-V1	$\mathcal{I} + \mathcal{S}$	33.9
L2G (CVPR22) [74]	R101-V2	$\mathcal{I} + \mathcal{S}$	44.2
CLIP-ES (CVPR23) [76]	R101-V2	$\mathcal{I} + \mathcal{L}$	45.4
CLIP-ES+CPAL (CVPR24) [77]	R101-V2	$\mathcal{I} + \mathcal{L}$	46.8
CONTA (NeurIPS20) [55]	WR38-V1	$\mathcal{I}$	32.8
CDA (ICCV21) [56]	WR38-V1	$\mathcal{I}$	33.2
AdvCAM (CVPR21) [27]	R101-V2	$\mathcal{I}$	44.4
SANCE (CVPR22) [83]	R101-V2	$\mathcal{I}$	44.7
SIPE (CVPR22) [25]	WR38-V1	$\mathcal{I}$	43.6
AMN (CVPR22) [9]	R101-V2	$\mathcal{I}$	44.7
MCTformer (CVPR22) [15]	WR38-V1	$\mathcal{I}$	42.0
OCR (CVPR23) [81]	WR38-V1	$\mathcal{I}$	42.5
BECO (CVPR23) [30]	R101-V3+	$\mathcal{I}$	45.1
LPCAM (CVPR23) [28]	WR38-V1	$\mathcal{I}$	42.8
CLIP-ES (CVPR23) [76]	R101-V2	$\mathcal{I}$	45.4
CTI (CVPR24) [62]	R101-V2	$\mathcal{I}$	45.4
AMN+CPAL (CVPR24) [77]	R101-V2	$\mathcal{I}$	46.3
MCTformer+ (TPAMI24) [16]	WR38-V1	$\mathcal{I}$	45.8
MCTTA-M (Ours)	R101-V3+	$\mathcal{I}$	49.4

$\mathcal{I}$: Image level label, $\mathcal{L}$: language label, $\mathcal{S}$: Saliency label.

TABLE IV
COMPARISONS OF THE mIoU (%) FOR SEGMENTATION RESULTS FOR VOC AND COCO FOR SINGLE-STAGE METHODS.

Single-Stage Methods	Backbone	VOC Val	VOC Test	COCO Val
RRM (AAAI20) [85]	WR38	62.6	62.9	-
1Stage (CVPR20) [33]	WR38	62.7	64.3	-
SLRNet (IJCV22) [86]	WR38	67.2	67.6	35.0
AFA (CVPR22) [63]	MIT-B1	66.0	66.3	38.9
ViT-PCM† (ECCV22) [87]	ViT-B†	70.3	70.9	-
ToCo (CVPR23) [36]	ViT-B	69.8	70.5	41.3
ToCo† (CVPR23) [36]	ViT-B†	71.1	72.2	42.3
MCTformer+D (TPAMI24) [16]	ViT-S	65.7	65.9	35.8
DuPL (CVPR24) [37]	ViT-B	73.3	72.8	44.6
MCTTA-D (Ours)	ViT-S	73.1	73.3	43.3
MCTTA-S (Ours)	ViT-S	75.4	75.2	45.8

† denotes using ImageNet21k [88] pretrained parameters.

and 80K on COCO.

4) *Direct Method Settings*: The direct method follows the same CAM generation process to produce raw CAMs, as described in Section IV-D3. Apart from CRFs, no additional refinement networks or techniques are applied.

C. Experimental Results

1) *Comparisons with Multi-Stage Methods*: Table I presents a comparison of mIoU (%) for raw CAMs (denoted as seed) and pseudo labels (denoted as mask) on the VOC and COCO across various multi-stage methods. The proposed MCTTA consistently outperforms existing approaches in both seed and mask evaluations, highlighting its effectiveness in CAM generation. On both VOC and COCO, MCTTA achieves the highest performance among all methods. Notably, compared to the strongest baseline, MCTformer+, our method improves CAM quality by more than 5.0 and enhances pseudo labels by 4.0 on VOC, demonstrating its superior ability to generate high-quality class activation maps and refined pseudo masks.

Table II and Table III compare the mIoU (%) for segmentation results on the VOC and COCO across various multi-stage methods, respectively. The proposed MCTTA-M achieves the state-of-the-art performance on both datasets. On VOC dataset, MCTTA-M achieves 77.8% mIoU on the val and 77.5% on the test, surpassing all existing methods. On COCO dataset, MCTTA-M achieves 49.4%, outperforming all prior methods. Compared to MCTformer+ (45.8%), our method provides a 3.6 improvement, demonstrating its robustness on large-scale datasets.

2) *Comparisons with Single-Stage Methods*: Table IV compares the mIoU (%) for segmentation results on VOC and COCO across various single-stage methods. The proposed MCTTA-S and MCTTA-D achieve the state-of-the-art performance. Specifically, despite using only ImageNet-1K pretrained weights, MCTTA-D performs competitively, surpassing ToCo [36], which utilizes a stronger ImageNet-21K [88] pretrained ViT-Base backbone. We also apply the direct method to MCTformer+ [16], resulting in MCTformer+-D. Although MCTformer+ excels in the multi-stage setting, it

AdamW optimizer with a batch size of 8 and a learning rate of 6×10^{-5} . MCTTA-D is trained for 20K iterations on VOC

TABLE V

COMPARISONS OF THE VANILLA AND PROPOSED SINGLE-STAGE METHOD.

Method	VOC Val	VOC Test	COCO Val
Vanilla	68.9	68.5	39.9
Vanilla+CRFs	70.2	70.3	41.9
Ours	74.8	74.4	44.8
Ours+CRFs	75.4	75.2	45.8

TABLE VI

ABLATION ON DIFFERENT OUTPUT STRIDES OF SPG.

Output Strides	Train	Val	Test
[8, 8, 16, 32]	73.4 (-0.5)	71.1 (-0.2)	71.2 (-0.2)
[8, 16, 16, 32]	73.1 (-0.8)	70.8 (-0.5)	71.0 (-0.4)
[8, 16, 32, 64]	73.9 (-0.0)	71.2 (-0.1)	71.4 (-0.0)
[16, 16, 16, 16]	72.5 (-1.4)	70.1 (-1.2)	70.3 (-1.1)
[16, 32, 32, 64]	73.2 (-0.7)	70.7 (-0.6)	70.9 (-0.5)
[16, 32, 64, 128]	73.0 (-0.9)	70.6 (-0.7)	70.6 (-0.8)
[16, 16, 32, 64]	73.9	71.3	71.4

struggles to achieve competitive performance when applied directly for segmentation.

Additionally, MCTTA-D achieves results comparable to DuPL [37], the strongest baseline, on both datasets. Notably, MCTTA-S, without incorporating additional losses [36] or training tricks [30], significantly outperforms DuPL. On VOC, MCTTA-S achieves 75.4% on val and 75.2% on test, surpassing all existing single-stage methods. Despite its simpler training process, MCTTA-S even surpasses the multi-stage MCTformer+, demonstrating its ability to generate high-quality pseudo labels. On COCO, MCTTA-S achieves 45.8%, outperforming previous methods, including DuPL (44.6%) and ToCo (42.3%), highlighting its robustness on large-scale datasets. MCTTA-D (44.0%) also performs competitively, further validating the effectiveness of our approach.

3) *Qualitative Comparisons of CAMs*: Fig. 10 (b), (d), and (f) show that the MCTTA series (direct, single-stage, and multi-stage) achieve more precise and sharper localization of object regions on VOC train set. In contrast, the CAMs from MCTformer+ (c), (e), and (g) exhibit less sharp localization of object boundaries and regions. These results highlight that MCTTA significantly improves upon MCTformer+ in boundary precision and fine-grained localization, demonstrating the effectiveness in WSSS.

4) *Qualitative Comparisons of Segmentation Results*: Fig. 11 shows segmentation results on the COCO val set across different methods. (b) MCTformer+-D exhibits inaccuracies and incomplete object boundaries, while (c) ToCo struggles with fine-grained details. (d), (e), and (f) show results from the proposed MCTTA series, which demonstrate improved object localization, where MCTTA-M provides the most accurate segmentation. Even the MCTTA direct method outperforms ToCo in object localization accuracy.

VI. ABLATION STUDIES

We evaluate the proposed single-stage method against the baseline and analyze the impact of key modules, including Spatial Prior Grapher (SPG), Class Token Projection (CTP),

TABLE VII

ABLATION ON DIFFERENT MODULES IN SPG.

Module	Train	Val	Test
ResNet Block	72.9 (-1.0)	70.0 (-1.3)	70.3 (-1.1)
ViT Block	73.1 (-0.8)	70.5 (-0.8)	70.4 (-1.0)
Grapher-Edge	73.5 (-0.4)	70.7 (-0.6)	71.0 (-0.4)
Grapher-GIN	73.6 (-0.3)	70.9 (-0.4)	71.0 (-0.4)
Grapher-SAGE	73.5 (-0.4)	70.8 (-0.5)	70.8 (-0.6)
Grapher-MR	73.9	71.3	71.4

TABLE VIII

ABLATION ON THE EFFECT OF DIFFERENT HIERARCHICAL LEVEL.

Level	Train	Val	Test
Remove All	69.8 (-4.1)	66.1 (-5.2)	66.6 (-4.8)
Remove H-1	72.1 (-1.8)	68.8 (-2.5)	68.9 (-2.5)
Remove H-2	71.7 (-2.2)	68.6 (-2.7)	68.9 (-2.5)
Remove H-3	72.8 (-1.1)	69.4 (-1.9)	69.4 (-2.0)
Remove H-4	72.8 (-1.1)	69.7 (-1.6)	69.0 (-1.4)
All Levels	73.9	71.3	71.4

Split Weighted Softmax, and the classification training weight λ . All experiments are conducted on the train, val, and test sets of VOC using MCTTA-D. Unless otherwise specified, results are reported without any refinement techniques, such as CRFs, to provide a clear assessment of the model's raw performance.

A. Effect of the Proposed Single-Stage Method

We compare the vanilla single-stage method and our proposed one in Table V. When under the same settings with CRFs, the proposed method consistently outperforms the vanilla method across all metrics. On VOC val, the proposed method achieves a 5.4 improvement. Similarly, on COCO val set, the proposed method outperforms the vanilla method by 3.9. This demonstrates the effectiveness of the proposed single-stage method.

B. Effect on Output Strides of SPG

Table VI presents an ablation on different output stride configurations of SPG to evaluate their impact on performance. The results indicate that the default stride setting [16, 16, 32, 64] achieves the best performance across the training, validation, and test sets.

Notably, non-hierarchical configuration, [16, 16, 16, 16] results in lower performance, likely due to the limitations of isotropic feature representation. Similarly, configurations with larger strides, such as [16, 32, 64, 128], also lead to performance degradation, suggesting that excessive downsampling may cause the loss of fine-grained details. When applying stride=[8, 16, 32, 64], the performance is almost the same as default setting. While the configuration [8, 16, 32, 64] performs comparably to the default setting, the use of stride = 8 significantly increases computational cost.

C. Effect on Modules in SPG

Table VII presents an ablation study on different module configurations within SPG. Following ViG [13], we evaluate

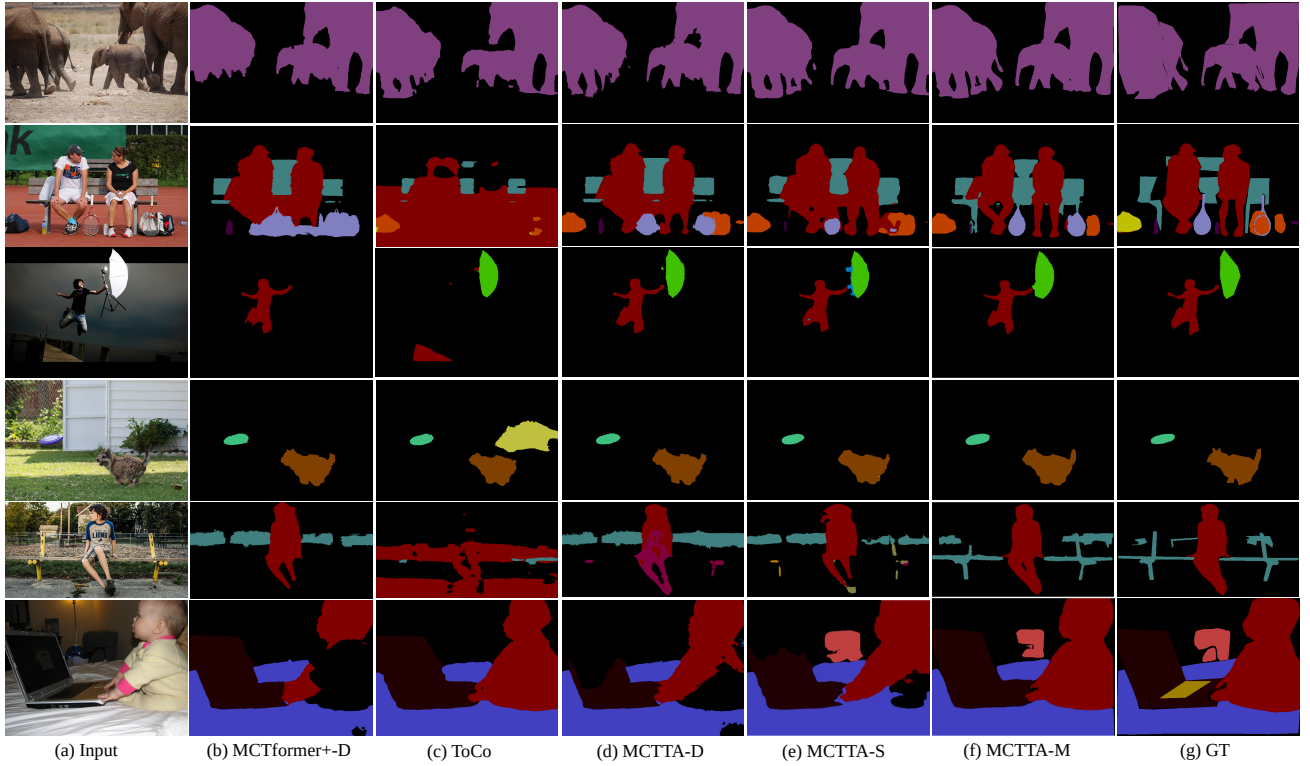


Fig. 11. Qualitative comparison of segmentation results across different methods on COCO val set [64]. (a): input image. (b): MCTformer+ [16] using direct method. (d-f): MCTTA direct, single-stage and multi-stage method. (g): ground truth.

TABLE IX
ABLATION ON THE EFFECT OF CLASS TOKEN PROJECTION.

Method	Train	Val	Test
None	71.7 (-2.2)	69.2 (-2.1)	69.4 (-2.0)
CTP- $\{4\}$	72.9 (-1.0)	70.1 (-1.2)	70.3 (-1.1)
CTP- $\{3,4\}$	73.0 (-0.9)	70.2 (-1.1)	70.1 (-1.3)
CTP- $\{2,3,4\}$	73.6 (-0.3)	70.7 (-0.6)	70.9 (-0.5)
CTP-All	73.9	71.3	71.4

TABLE X
ABLATION ON DIFFERENT WEIGHTS FOR SPLIT WEIGHTED SOFTMAX.

(α_1, α_2)	Train	Val	Test
Vanilla	70.6 (-3.3)	67.4 (-3.9)	67.8 (-3.6)
(0.25, 0.75)	72.8 (-1.1)	69.5 (-1.8)	69.6 (-1.8)
(0.50, 0.50)	73.4 (-0.5)	70.7 (-0.6)	70.9 (-0.5)
(0.75, 0.25)	73.1 (-0.8)	70.5 (-0.8)	70.5 (-0.9)
(1.50, 1.00)	72.1 (-1.8)	69.4 (-1.9)	69.4 (-2.0)
(1.00, 1.50)	71.6 (-2.3)	68.8 (-2.5)	68.9 (-2.5)
(1.00, 1.00)	73.9	71.3	71.4

various graph convolution settings, including Edge [89], GIN [90], SAGE [91], and Max-Relative (MR) [92]. Additionally, we incorporate ResNet and ViT blocks to assess the impact of different architectures for processing hierarchical tokens.

Among all configurations, the graph-based modules perform competitively, with Grapher-MR achieving the best performance across all metrics, consistent with findings in ViG. In contrast, the ResNet Block and ViT Block show the

largest performance drop, indicating that conventional CNN and Transformer blocks alone are insufficient for capturing spatial dependencies in WSSS.

D. Effect on Hierarchical Levels

Table VIII presents an ablation study on the impact of different hierarchical levels (denoted as H- i). The results show that removing any hierarchical level results in a significant decrease in performance. Specifically, removing all hierarchical levels leads to a substantial performance drop, emphasizing the importance of incorporating all hierarchical features. Additionally, the removal of any individual level, particularly H-1, causes notable performance degradation.

E. Effect on Class Token Projection

Table IX presents an ablation on the effect of CTP. “None” indicates that the model does not use CTP, while “CTP- $\{i\}$ ” denotes that the CTP uses the i -th level for generating the class token. The results show that CTP-All, the full configuration, achieves the best performance. Among the partial CTP configurations, CTP- $\{2,3,4\}$ exhibits the smallest performance drop. Removing CTP entirely leads to a significant performance decline, highlighting the importance of input-aware class token learning.

F. Effect on Split Weighted Softmax

Table X presents an ablation on the impact of different weight configurations (α_1, α_2) for Split Weighted Softmax.

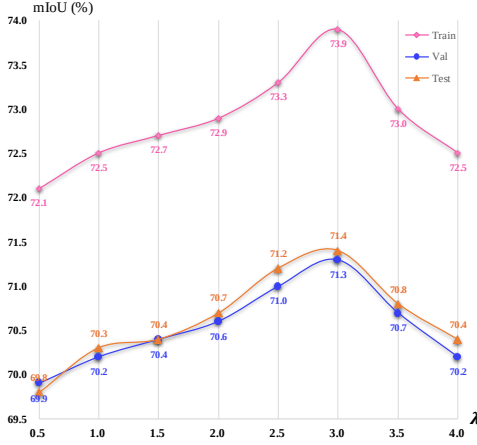


Fig. 12. Evaluation on the effect of λ for classification training.

“Vanilla” denotes that the traditional softmax is applied in attention. The results indicate that the default weight setting (1.00, 1.00) achieves the best performance across all metrics, where the vanilla settings greatly affects the performances.

When weights are adjusted, the configuration (0.50, 0.50) yields the second-best performance, with only a slight decrease, as it treats the two attention scores $\mathbf{A}_c$ and $\mathbf{A}_p$ equally. In contrast, the imbalanced configurations (0.25, 0.75) and (0.75, 0.25) result in more significant performance drops, particularly on the validation and test sets. Additionally, we observe that setting either α_1 or α_2 greater than 1 leads to a substantial decrease in performance.

G. Effect on Weight λ for Classification Training

Fig. 12 shows the evaluation of the effect of the weight λ on the performance during classification training. As observed, performance improves with increasing λ up to a point, reaching its peak at $\lambda = 3.0$. Beyond this point, further increasing λ leads to a decline in performance, particularly on the test set, which drops to 70.4% at $\lambda = 4$.

This suggests that a balance in the weight λ is crucial for optimizing classification performance, and excessively high values may lead to overfitting or suboptimal generalization.

VII. CONCLUSION

In this paper, we introduce MCTTA, a novel WSSS framework that enhances the MCTformer+ baseline by incorporating graph-based multi-scale feature extraction and a hierarchical design. Our approach effectively improves quality of the object localization map and CAM. To achieve this, we propose key modules, including Spatial Prior Grapher (SPG), Class Token Projection (CTP), Cross Attention Block, and Split Weighted Softmax, which together enhance feature representation and refine the segmentation results. Extensive experiments on PASCAL VOC 2012 and MS COCO 2014 demonstrate the superiority of MCTTA over existing methods. Specifically, MCTTA-D and MCTTA-S achieve the state-of-the-art performance among single-stage methods, while MCTTA-M surpasses multi-stage approaches, proving its robustness

across different datasets. Furthermore, comprehensive ablation studies validate the effectiveness of each proposed module, consistently showing performance gains across various configurations. These findings highlight the flexibility, scalability, and effectiveness of MCTTA, setting a strong foundation for future advancements in WSSS.

In future work, we will focus on the challenge of effectively leveraging GNN to capture and exploit nuanced class relationships for enhancement.

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Wataru KAMEYAMA received Bachelor, Master and Doctor of Eng. from School of Sc. and Eng., Waseda Univ. in 1985, 1987 and 1990, respectively. He joined ASCII Corp. in 1992, and was transferred to France Telecom CCETT from 1994 to 1996 for his secondment. After joining Waseda Univ., as an associate professor in 1999, he has been appointed as a professor at Department of Communications and Computer Engineering, School of Fundamental Sc. and Eng., Waseda Univ. since 2014. He has been involved in MPEG, MHEG, DAVIC and the TV-Anytime Forum activities. He was a chairman of ISO/IEC JTC1/SC29/WG12, and a secretariat and a vice chairman of the TV-Anytime Forum. He is a senior member of IEICE and IPSJ, and a member of ITE, IEEEJ, ACM and IEEE.



Dingjie PENG received the B.S. degree in Computer Science from Shenzhen University (Shenzhen, China) in 2021 and received the M.S. degree in Computer Science and Communications Engineering from Waseda University (Tokyo, Japan) in 2023. Since 2023, he is a Ph.D. candidate at Waseda University. His research interests mainly focus on computer vision and deep generative artificial intelligence.